

Fractional extremizers force near-design cores

Research continuation, 7 October 2026. We prove quantitative deletion stability and an asymptotic if-and-only-if characterization at the integer edge/rank ratios. The finite structural result uses no linearity or small-intersection assumption. Its integer-cover corollary uses an explicitly stated theorem of Kahn.

1. Conventions and statements

Let H be a finite simple intersecting r -uniform hypergraph, $r \geq 2$, with $m \geq 1$ edges. Vertices outside all edges are ignored. A vertex cover meets every edge, and its minimum size is $\tau(H)$. A fractional vertex cover has nonnegative vertex weights of sum at least one on each edge; its minimum total is $\tau^*(H)$. Equivalently, by finite LP duality, $\tau^*(H)$ is the maximum of $Y = \sum_E y_E$ over nonnegative edge weights satisfying

$$\sum_{E \ni v} y_E \leq 1 \quad \text{for every vertex } v. \quad (1)$$

The optimum exists. Appendix A records the duality justification.

For any intersecting subfamily J put

$$I(J) = \sum_{\{E, F\} \subset E(J)} (|E \cap F| - 1). \quad (2)$$

Pairs are unordered and distinct. Set $I = 0$ for an empty subfamily. Every summand is nonnegative; $I = 0$ means linearity.

Theorem 1 (general finite extraction). Fix an integer $k \geq 2$ and any feasible weights (1). If $Y > m/(k+1)$, retain exactly the edges with $y_E > 1/(k+1)$. Let J be this core, q its edge count, and $s = m - q$ the number deleted. Then every active vertex of J has degree at most k and

$$s \leq \frac{m(k+1)\{mr - (m+r-1)Y\}}{(r-1)\{(k+1)Y - m\}}. \quad (3)$$

Both factors in the denominator are positive. Independently of the mass condition, this same threshold core has maximum degree at most k . It satisfies

$$\sum_v d_J(v)(k - d_J(v)) = q\{(k-1)r - q + 1\} - 2I(J) \geq 0, \quad (4)$$

including $q = 0$. In particular $I(J) \leq q\{(k-1)r - q + 1\}/2$.

Theorem 2 (linear deletion stability at a design boundary). Suppose $m = (k-1)r + 1$, $k \geq 2$, and put

$$d = m/k - \tau^*(H) \geq 0. \quad (5)$$

There is a core J , obtained by the threshold above from any optimal dual vector, for which

$$s \leq \frac{2k^2(k+1)r}{r-1} d \leq 4k^2(k+1)d, \quad \max_v d_J(v) \leq k, \quad I(J) \leq qs/2. \quad (6)$$

If n counts its active vertices, then

$$\sum_v d_J(v)(k - d_J(v)) = qs - 2I(J), \quad 0 \leq kn - qr \leq qs. \quad (7)$$

The dependence on d cannot be replaced by $o(d)$ uniformly, already for $k = 2$; Section 5 gives a family whose minimum deletion count is $2d$. No optimality of the coefficient in (6) is claimed.

Theorem 3 (complete asymptotic characterization at integer ratios). Fix a positive integer a and a sequence of such hypergraphs with $r_j \rightarrow \infty$ and $m_j/r_j \rightarrow a$. The following are equivalent:

1. $\tau^*(H_j)/r_j \rightarrow a/(a+1)$.
2. Deleting $o(r_j)$ edges leaves a core with maximum vertex degree at most $a+1$.

For the cores in either condition, $q_j/r_j \rightarrow a$,

$$I(J_j) = o(r_j^2), \quad n_j/r_j^2 \rightarrow a/(a+1), \quad \sum_v (d_{J_j}(v) - a - 1)^2 = o(r_j^2). \quad (8)$$

Moreover

$$\tau(H_j)/r_j \rightarrow a/(a+1), \quad \tau(H_j)/\tau^*(H_j) \rightarrow 1. \quad (9)$$

Only (9) uses the published rounding input stated in Section 4. The hypotheses impose no bound on intersections in the whole family. In fact (8) can fail for the whole family even when (9) holds exactly, as Section 5 demonstrates.

2. The weighted estimate and finite extraction

Choose an edge P of maximum weight b . Every other edge meets P , so nonnegative weights and (1) give

$$Y - b \leq \sum_{v \in P} \sum_{E \neq P, v \in E} y_E \leq r(1 - b). \quad \text{Thus } Y \leq r - (r - 1)b. \quad (10)$$

Multiple intersections only increase the middle sum. Since $Y \leq mb$,

$$Y \leq \frac{mr}{m + r - 1}, \quad (r - 1)(mb - Y) \leq mr - (m + r - 1)Y =: \Psi. \quad (11)$$

In particular $\Psi \geq 0$. These facts hold for every feasible vector.

For a deleted edge, $b - y_E \geq b - 1/(k + 1)$. Thus

$$s\{b - 1/(k + 1)\} \leq \sum_{E \text{ deleted}} (b - y_E) \leq mb - Y. \quad (12)$$

Here all peak deficits $b - y_E$ are nonnegative. Since $b \geq Y/m$ and $Y/m > 1/(k + 1)$, combining (11)–(12) proves (3).

If a vertex belonged to at least $k+1$ retained edges, their weights would sum to more than one. Hence the retained degree is at most k . For the core, double counting incidences and edge-pair intersections gives

$$\sum_v d_J(v) = qr, \quad \sum_v \binom{d_J(v)}{2} = \binom{q}{2} + I(J).$$

Consequently $\sum_v d_J(v)^2 = qr + q(q - 1) + 2I(J)$. Subtracting this from kqr proves (4); all summands on its left are nonnegative. This proves Theorem 1, including empty cores.

3. The design boundary and the iff statement

At $m = (k - 1)r + 1$, equation (11) gives $\tau^* \leq m/k$, proving $d \geq 0$. For an optimal dual vector, $Y = m/k - d$ and $\Psi = krd$. If $d \leq m/[2k(k + 1)]$, then

$$(k + 1)Y - m = m/k - (k + 1)d \geq m/(2k) > 0.$$

Insert these estimates into (3) to obtain the first bound in (6). If d is larger instead, the trivial bound $s \leq m$ gives $s \leq 2k(k + 1)d$, which is stronger. This proves (6) for every d , with no small-deficit hypothesis. Formula (4) becomes the first identity of (7). For each active degree $1 \leq d_J(v) \leq k$, $0 \leq k - d_J(v) \leq d_J(v)(k - d_J(v))$. Summing proves the second claim of (7). This completes Theorem 2.

For Theorem 3 put $k = a + 1$. Under condition 1,

$$\Psi_j/r_j^2 = m_j/r_j - (m_j/r_j + 1 - 1/r_j)Y_j/r_j \rightarrow 0.$$

Also $Y_j/m_j \rightarrow 1/k > 1/(k + 1)$. Applying (3) shows $s_j/r_j \rightarrow 0$. This proves condition 2, with an actual threshold core.

Conversely, if a core of size $q_j = m_j - o(r_j)$ has degree at most k , put weight $1/k$ on each core edge and zero on each deleted edge. These weights are feasible for the original family, giving $\tau^*(H_j) \geq q_j/k$. The harmonic upper bound (11) proves condition 1. This argument is valid for any such core.

For any core as in condition 2, (4) implies

$$0 \leq \sum_v d_J(v)(k - d_J(v)) \leq q\{(k - 1)r - q + 1\} = o(r^2), \quad 0 \leq 2I(J) \leq q\{(k - 1)r - q + 1\} = o(r^2).$$

The bracket is nonnegative for nonempty cores, by (4). Summing $k - d_J$ as above shows $kn - qr = o(r^2)$. Finally, for $1 \leq d_J \leq k$, $(k - d_J)^2 \leq (k - 1)d_J(k - d_J)$; summation proves the variance assertion (8). This proves every structural part of Theorem 3 without a rounding theorem.

4. Integer rounding and what is imported

Input K (Kahn 1994, Corollary 5.4, printed p.140). For fixed $C > 0$, an intersecting r -uniform sequence with at most Cr edges and maximum intersection of distinct edges $o(r)$ satisfies

$$\tau/r \leq C/(C + 1) + o(1). \tag{13}$$

This is a published input, not an assertion proved or formalized in this note. We explain precisely how its hypotheses are reached.

Let J be a core from Theorem 3 and write $\epsilon = I(J)/r^2 \rightarrow 0$. Put $\rho = \sqrt{\epsilon} + r^{-1/2}$, so $\rho \rightarrow 0$ and $\rho r \geq \sqrt{r}$. Every edge pair with intersection at least ρr contributes at least $\rho r - 1$ to $I(J)$. The number of such pairs is at most

$$I(J)/(\rho r - 1) = o(r).$$

Indeed its ratio to r is $\epsilon/(\rho - 1/r) \rightarrow 0$. Choose and delete one edge from each such pair. At most $o(r)$ edges are deleted, and no bad pair remains. The resulting family J' has maximum distinct-edge intersection $< \rho r = o(r)$.

For each fixed $\eta > 0$, J' has at most $(a + \eta)r$ edges eventually. Applying Input K and then letting η decrease to zero gives $\limsup \tau(J')/r \leq a/(a + 1)$. Each deleted edge, from either stage, can be

covered with one of its r vertices; thus the total extra cost is $o(r)$. Therefore the same upper bound holds for H . The lower bound $\tau(H) \geq \tau^*(H)$ proves (9).

One consequence is a uniform separation statement: for each fixed integer a and $\eta > 0$ there exist $\delta > 0$ and R such that, if $r \geq R$, $|m/r - a| \leq \delta$ and $\tau/r \geq a/(a+1) + \eta$, then $\tau^*/r \leq a/(a+1) - \delta$. If this failed, for $\delta = 1/j$ and $R = j$ one could choose a violating family with $m/r \rightarrow a$, integer cover ratio bounded above the endpoint, and fractional ratio tending to the endpoint by (11). This contradicts Theorem 3. No explicit value of δ , no rate in Input K, and no resolution of unrestricted Ryser's conjecture are asserted.

5. Sharp deletion order and unavoidable exceptional edges

Here is a complete elementary family for $k = 2$. Fix even $n \geq 4$ and an integer $t \geq 1$, and put $r = n - 1 + t$. For each pair of distinct indices i, j in $\{1, \dots, n\}$ use a vertex $x_{ij} = x_{ji}$. Define n good edges

$$A_i = \{x_{ij} : j \neq i\} \cup \{u_{i,1}, \dots, u_{i,t}\},$$

where all padding vertices u are distinct and fresh. Thus $|A_i| = r$, and distinct good edges meet exactly once. Let C be the $n/2$ vertices of one fixed perfect matching of the index set. Each good edge contains exactly one vertex of C .

Take a fresh common set W of size $r - n/2 - 1 = n/2 + t - 2$, and fresh distinct vertices $p_1, \dots, p_t$. The t bad edges are

$$B_t = C \cup W \cup \{p_t\}.$$

Each has r vertices; all edges are distinct. Each bad edge meets each good edge exactly once, and two bad edges meet in $r-1$ vertices. The whole family is intersecting, with $m = n + t = r + 1$.

Put dual weight $1/2$ on the good edges and zero on the bad edges. Pair vertices have incident dual weight one, good private vertices have weight $1/2$, and all other vertices have weight zero. This is feasible with total $n/2$. The integer cover C has size $n/2$ and meets every edge. Weak duality therefore certifies

$$\tau = \tau^* = n/2, \quad d = m/2 - n/2 = t/2. \quad (14)$$

Deleting all bad edges leaves degree at most two. Conversely, let ℓ bad edges remain in any degree-at-most-two subfamily. At a vertex of C they already have degree ℓ , so $\ell \leq 2$. Each of the $n/2$ disjoint matching pairs originally contributed two good-edge incidences at its vertex; at least ℓ of these two good edges must be deleted. These requirements involve disjoint good-edge sets, so at least $\ell n/2$ good edges are deleted. The total deletion count is consequently at least

$$t - \ell + \ell n/2 \geq t.$$

Its exact minimum is $t = 2d$. This proves the sharp linear order claim.

Only bad-bad pairs contribute excess, so

$$I(H) = \binom{t}{2}(r-2). \quad (15)$$

For the exact integer sequence $n = 2j^4$ and $t = j^3$, $j \geq 2$, we have $t = o(r)$, $m/r \rightarrow 1$ and fractional ratio tending to $1/2$, but $I(H)/r^2 \rightarrow \infty$. Even the integer ratio agrees with its fractional counterpart exactly. Thus Theorem 3 cannot require the whole original family to have $I = o(r^2)$, nor can it omit deletion.

6. Verification scope and significance limits

The finite extraction inequality, threshold degree cap, and uniform-weight converse are formalized for actual finite incidence sets in the accompanying pinned Lean project. The weighted-star proof sums over real edge and vertex sets and uses the genuine uniformity, intersectingness and feasibility hypotheses; these are not replaced by a scalar assumption. Kernel reports contain only `propext`, `Classical.choice` and `Quot.sound`.

The degree/excess double counting, real-sequence limit arguments, LP duality, the sharpness construction and Input K are written proofs, not full-paper Lean formalizations. Exact rational diagnostics check incidence identities, the extraction bound and explicit primal/dual sharpness certificates; they are not substitutes for the universal proofs above. No external human review, historical priority or high-level prize significance is claimed.

Appendix A. The finite duality fact

Let A be the vertex-by-edge incidence matrix. Weak duality follows by summing $\sum_{E \ni v} y_E \leq 1$ against any feasible primal weights. The primal infimum α is attained after clipping each vertex weight to at most one, which preserves feasibility; the dual is compact because every edge is nonempty and hence each $y_E \leq 1$.

For completeness use the finitely generated closed cone

$$\mathcal{C} = \{(A^T z - s, \mathbf{1}^T z + t) : z, s, t \geq 0\} \subset \mathbb{R}^{m+1}.$$

For $\beta < \alpha$, the point $(\mathbf{1}, \beta)$ is outside it. Finite-dimensional separation yields (u, b) , nonnegative on this cone and strictly negative on that point. The generators $(-e_i, 0)$ and $(0, 1)$ give $u \leq 0$ and $b \geq 0$. With $y = -u$, vertex generators imply $Ay \leq b\mathbf{1}$, while strict separation gives $\sum_i y_i > b\beta$. If $b = 0$, nonempty edges and $y \geq 0$ would imply $y = 0$, a contradiction. Thus y/b is a feasible dual vector of value $> \beta$. Letting β increase to α and using dual compactness proves equality of optima.

The standard finite convex separation theorem is used here. Closedness of the finitely generated cone follows by choosing a representation with linearly independent generators, deleting dependencies while preserving nonnegative coefficients. Along a convergent sequence pass to one fixed independent subset of the finitely many generators; its coefficients then converge by the inverse on their linear span. This justifies the needed closedness.

References and provenance

[K] J. Kahn, *On a Problem of Erdős and Lovász. II: $n(r)=O(r)$* , J. Amer. Math. Soc. **7** (1994), 125–143. DOI <https://doi.org/10.1090/S0894-0347-1994-1224593-5>, Corollary 5.4. The harmonic integer bound and rounding theorem are prior work.

[F] Repository note *A universal fractional cover envelope and its design equality cases*, commit 4ae9051, 7 October 2026. Its stronger noninteger envelope and exact equality classification are preserved. The present deletion estimate uses only the weighted-star inequality and supplies a separate quantitative stability theorem.

The limited literature screen also identifies Kayll’s DIMACS Report 95-55, Theorem 2.1, as an existing resolution of Kahn’s Conjecture 5.6. That theorem is not an input here. The new finite extraction and fixed-ratio characterization remain subject to further novelty comparison; no first-result wording is used.